(a) 1, 3 and 4 (b) 2, 3, 4 and 5
(c) 2, 4 and 5 (d) 1, 2, 3 and 5

Ans. (c) : Conservation Agriculture (CA) can be defined by a statement given by the Food and Agricultural Organization of the United Nation as "a concept for resource-saving agricultural crop production that strives to achieve acceptable profits together with high and sustained production levels while concurrently conserving the environment". The 3 principle of CA are:
(i) Minimum tillage and soil disturbance.
(ii) Permanent soil cover with crop residues and live mulches.
(iii) Crop rotation and intercropping.

66. With reference to the Genetically Modified mustard (GM mustard) developed in India, consider the following statements:
1. GM mustard has the genes of a soil bacterium that give the plant the property of pest-resistance to a wide variety of pests.
2. GM mustard has the genes that allow the plant cross-pollination and hybridization.
3. GM mustard has been developed jointly by the IARI and Punjab Agricultural University.
Which of the statements given above is/are correct?
(a) 1 and 3 only (b) 2 only
(c) 2 and 3 only (d) 1, 2 and 3

Ans. (b): DMH -11 is a Genetically Modified (GM) mustard hybrid. Hybrids are normally obtained by crossing two genetically diverse plants from the same species. The 1st generation offspring resulting from it has higher yields than what either of the parents is individually capable of giving. But there is no natural hybridization system in mustard, unlike in rice, cotton, maize or tomato. This is because its flowers contain both the female (pistil) and male (stamen) reproductive organs, making the plant naturally self-pollinating. So the correct answer is option (b).

COMPUTERS

67. The terms 'WannaCry, Petya and EternalBlue' sometimes mentioned in the news recently are related to
(a) Exo-planets (b) Crypto-currency
(c) Cyber attacks (d) Mini satellites

Ans. (c): The Petya and Wanna Cry cyber-attacks in May and June 2017; are two of the biggest in history and impacted the finances of companies throughout the globe. Wanna Cry, which affected large numbers of organization, including the NHS, spread to 150 countries and is estimated to have cost the global economy 6 billion euro.
Eternal Blue an exploit developed by the U.S. National Security Agency (NSA) in an older windows systems released by the Shadow Brokers and was used as part of the worldwide Wannacry Ransomware attack.

Science & Technology

68. What is/are the consequence/consequences of a country becoming the member of the 'Nuclear Suppliers Group'?
1. It will have access to the latest and most efficient nuclear technologies.
2. It automatically becomes a member of "The Treaty on the Non-Proliferation of Nuclear Weapons (NPT)".
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : The Nuclear Supplier Group (NSG) is a group of nuclear supplier countries that seeks to contribute to the non-proliferation of nuclear weapons through the implementation of two sets of guidelines for nuclear exports and nuclear- related exports. But becoming a member of the NSG will not automatically make a country a member of the treaty on the Non-proliferation of Nuclear Weapons (NPT).
The NSG is a multilateral export control regime and a group of nuclear supplier countries that seek to prevent nuclear proliferation by controlling the export of materials, equipment and technology that can be used to manufacture nuclear weapons.
Membership of the NSG enables access to technology for a range of uses, from medicine to building nuclear power plants for India from the NSG, which is essentially a traders cartel.
So, the correct answer is option (a).

69. Consider the following statements:
1. Aadhaar card can be used as a proof of citizenship or domicile.
2. Once issued, Aadhaar number cannot be deactivated or omitted by the Issuing Authority.
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d): The Aadhaar (Targeted Delivery of Financial and Other Subsidies, Benefits and Services) Bill 2016 says that the number can't be used as proof of citizenship or domicile.
Statement (1) is not correct.
- The Aadhaar Act regulations state that an individual Aadhaar number may be "omitted" permanently or deactivated temporarily by the Unique Identification Authority of India, the agency responsible for issuing the numbers and managing the database. The regulations give the authority the power to deactivate Aadhaar numbers even in the absence of an effective grievance redressed procedure for those whose numbers have been suspended.

Statement (2) is not correct.

70. The identity platform 'Aadhaar' provides open "Application Programming Interfaces (APIs)". What does it imply?
1. It can be integrated into any electronic device.
2. Online authentication using iris is possible.

Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : The Aadhaar Act regulations, which are currently in parliament, state that an individual's Aadhaar number may be "omitted" permanently or deactivated temporarily by the Unique Identification Authority of India, the agency responsible for issuing the numbers and managing the database. The regulations gave the Authority the power to deactivate Aadhaar numbers even in the absence of an effective grievance redressal procedure for those whose numbers have been suspended. Calcutta High Court on 26th December, 2016 ruled that Aadhaar card is not proof of citizenship and stated that "Aadhaar Card by itself should not confer any right of or be proof of citizenship or domicile in respect of the holder there to" while rejecting a claim of citizenship of the accused who was issued the card in view of his long residence in the country.
Hence, the correct answer is option (d)

71. In the Indian context, what is the implication of ratifying the 'Additional Protocol' with the 'International Atomic Energy Agency (IAEA)'?
(a) The civilian nuclear reactors come under IAEA safeguards.
(b) The military nuclear installations come under the inspection of IAEA.
(c) The country will have the privilege to buy uranium from the Nuclear Suppliers Group (NSG).
(d) The country automatically becomes a member of the NSG.

Ans. (a) : India has ratified the Additional Protocol, a commitment given under India-US nuclear deal by the previous dispensation, to grant greater ease to International Atomic Energy Agency to monitor India's civilian atomic programme. The IAEA had in March 2009 approved an additional protocol to India's safeguards agreement consequent to a pact reached with the agency the previous year to place its civilian nuclear facilities under IAEA safeguards. That agreement had paved the way for the 45-number Nuclear Suppliers Group to grant India-specific waiver for it to have commercial relations with other countries in the civilian atomic field.

72. Which of the following is/are the aim/aims of "Digital India" Plan of the Government of India ?
1. Formation of India's own Internet companies like China did.
2. Establish a policy framework to encourage overseas multinational corporations that collect Big Data to build their large data centres within our national geographical boundaries.
3. Connect many of our villages to the Internet and bring Wi-Fi to many of our schools, public places and major tourist centres.
Select the correct answer using the code given below :
(a) 1 and 2 only (b) 3 only
(c) 2 and 3 only (d) 1, 2 and 3

Ans. (b): In order to transform the entire ecosystem of public services through the use of information technology, the Government of India has launched the Digital India programme with the vision to transform India into a digitally empowered society and knowledge economy. Under the Digital India Plan, NII would integrate the network and cloud infrastructure in the country to provide high-speed connectivity and platform to various government departments up to the panchayat level. Cities with a population of over 1 million and tourist centres could be provided with public wi-fi hotspots to promote digital cities. The scheme would be implemented by DoT and the Ministry of Urban Development (MoUD). Hence only statement (3) is correct.

73. What is "Terminal High Altitude Area Defence (THAAD)", sometimes seen in the news ?
(a) An Israeli radar system
(b) India's indigenous anti-missile programme
(c) An American anti-missile system
(d) A defence collaboration between Japan and South Korea

Ans. (c) : Terminal High Altitude Area Defense (THAAD), formerly Theater High Altitude Area Defence, is an American anti-ballistic missile defense system designed to shoot down short, medium and intermediate-range ballistic missiles in their terminal phase (descent or recently) by intercepting with a hit-to-kill approach.

74. In which of the following areas can GPS technology be used ?
1. Mobile phone operations
2. Banking operations
3. Controlling the power grids
Select the correct answer using the code given below:
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d): Global Positioning System (GPS) applications generally fall into 5 major categories-
(i) Location - determining a position
(ii) Navigation - getting from one location to another
(iii) Tracking- monitoring object or personal movement.
(iv) Mapping - creating maps of the world.
(v) Timing - bringing precise timing to the world.
GPS's accurate time facilities in everyday activities such as banking, mobile phone operations and even the control power grids by allowing well synchronized hand-off switching.

75. With reference to the Indian Regional Navigation Satellite System (IRNSS), consider the following statements :
1. IRNSS has three satellites in geostationary and four satellites in geosynchronous orbits.
2. IRNSS covers entire India and about 5500 sq. km area beyond its borders.
3. India will have its own satellite navigation system with full global coverage by the middle of 2019.
Which of the statements given above is/are correct ?
(a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) None